Instructions to ECE Students for their Electives

(From 2023 batch onwards)

July 2025

BTech Regular: By the time of graduation, BTech ECE students must do 20 credits of ECE Electives in the following manner:

- One star elective from Communications (4 credits) and one star elective from Electronics (4 credits): 8 credits in total
- Two two-credit courses from two different streams* in 4th semester: 4 credits in total
- Eight credits of electives from any streams* (with a course number of 300) and above: 8 credits in total

BTech Honours: By the time of graduation, BTech ECE students must do 20 credits of ECE Electives in the following manner:

- One star elective from communication (4 credits) and one star elective from Electronics (4 credits): 8 credits in total
- Two two-credit courses from two different streams in 4th semester*: 4 credits in total
- 12 Credits of stream electives (with a course number of 300 and above) from any ONE of the three streams*. Open Elective slots may also be used for these electives by obtaining consent from the respective Honours Advisor.

BTech Dual Degree: By the time of graduation, BTech ECE students must do 32 credits of ECE Electives in the following manner:

- One star elective from communication (4 credits) and one star elective from Electronics (4 credits): 8 credits in total
- Two two-credit courses from two different streams* in the 4th semester: 4 credits in total
- 20 credits of stream electives (with a course number of 300 and above) from any ONE of three streams.

*** <u>Streams allowed: VLSI and Embedded Systems, SPC, Robotics.</u>**

Note that Honours and DD students, who are not working in these three centres, must choose a stream from one of these three streams.

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Star Elective Courses in Electronics (4 credits): -

Analog IC Design
Digital VLSI Design
Mechatronic System Design

Star Elective Courses in Communications (4 credits): -

Wireless Communications
Signal Detection and Estimation Theory
Information Theory
Modern Coding Theory

A list of streams, along with foundation/stream electives, is given on the next page:

VLSI and Embedded Systems Stream

Two-credit Foundation Courses (offered only in 4th semester)

Radio Frequency Based Sensors Design: Principles and Applications

Stream Electives

Analog IC Design
Digital VLSI Design
Advanced Devices
Design for Testability
VLSI Architectures
CMOS References and Regulators
Hardware for AI
Principles of Semiconductor Devices
Advanced Computer Architecture
Bio-instrumentation and Devices
Flexible Electronics
Statistical Methods in AI

Signal Processing and Communications Stream

Two-credit Foundation Courses (offered only in 4th semester)

Introduction to Coding Theory
Communications and Controls in IoT* (Not being offered this year)
Introduction to Statistical Signal Processing

Stream Electives

Wireless Communications
Information Theory
Modern Coding Theory
Adaptive Signal Processing
Signal Detection and Estimation Theory
Speech Signal Processing
Medical Image Processing
Topics in Information-Theoretic Privacy
Information Theory
Topics in Deep Learning
Quantum Error Correcting Codes
Statistical Methods in AI
Modern Coding Theory
Time Frequency Analysis
Optimization Methods
Digital Image Processing
Computer Vision
Topics in Coding Theory

Robotics Stream

Two-credit Foundation Courses (offered only in 4th semester)

Introduction to Robotic Perception Planning and Control

Stream Electives

Robotics: Dynamics and Control
Introduction to UAV Design
Mechatronic System Design
Mobile Robotics
Statistical Methods in AI
Robotics: Planning and Navigation
Digital Image Processing
Topics in ML
Computer Vision
Topics in Deep Learning
Advances in Robotics & Control

Category labels for the above courses are given below:

For Star electives – **EC-Star-Elec**; **EC-Star-Comm**
For 2-Cr foundation courses – **EC-Found-VLSI; EC-Found-Robotics, EC-Found-SPC**
For Stream electives – **EC-Stream-VLSI, EC-Stream-Robotics, EC-Stream-SPC**

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